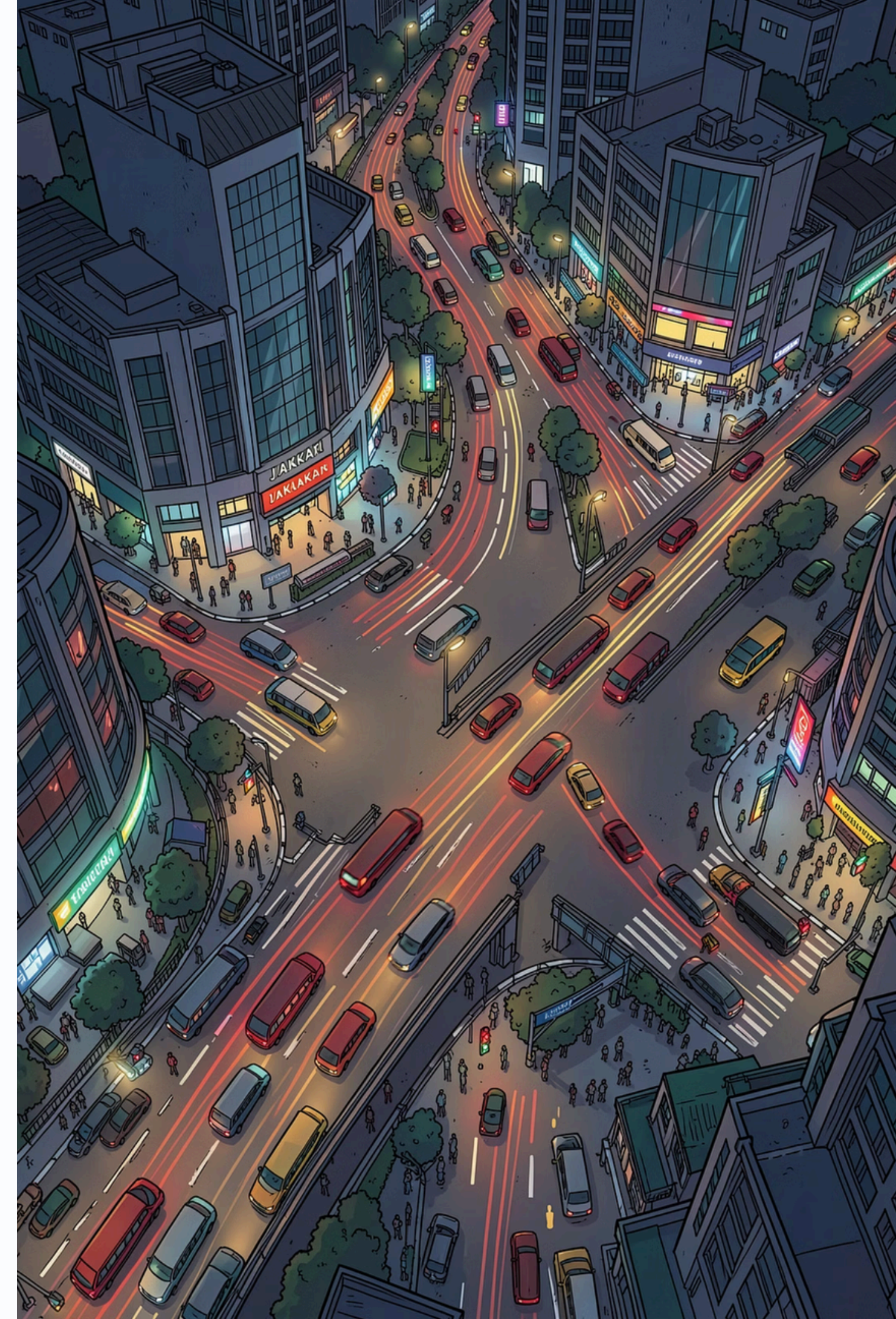


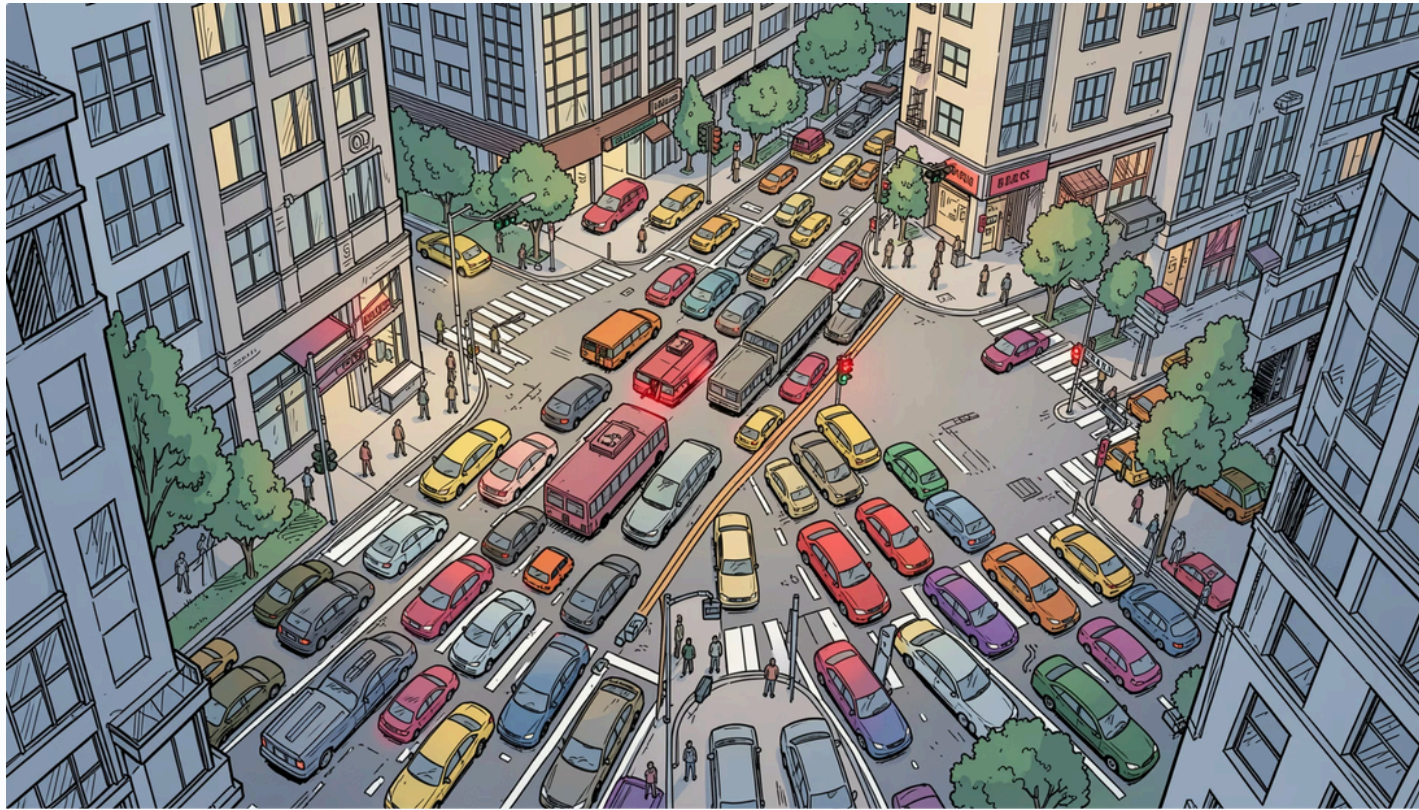


Adaptive Traffic Signal Optimization Using YOLOv26 Object Detection and PCE-Weighted Fuzzy Logic

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The Urban Traffic Crisis



Rapid urbanization and population growth have overwhelmed traditional infrastructure. **Fixed-time traffic signals** ignore real-time vehicle density, causing:

Economic Loss

Wasted fuel and productivity from idling vehicles

Environmental Damage

Increased CO₂ emissions from unnecessary stops

Driver Frustration

Green lights on empty lanes while others queue

What Exists & What's Missing

YOLO Frameworks

YOLOv8 is widely studied but faces **occlusion problems** — larger vehicles hide smaller ones in heavy congestion, rain, or fog.

Fuzzy Logic

Translates vehicle counts into **low/medium/high density** categories, enabling robust timing decisions even with detection noise.

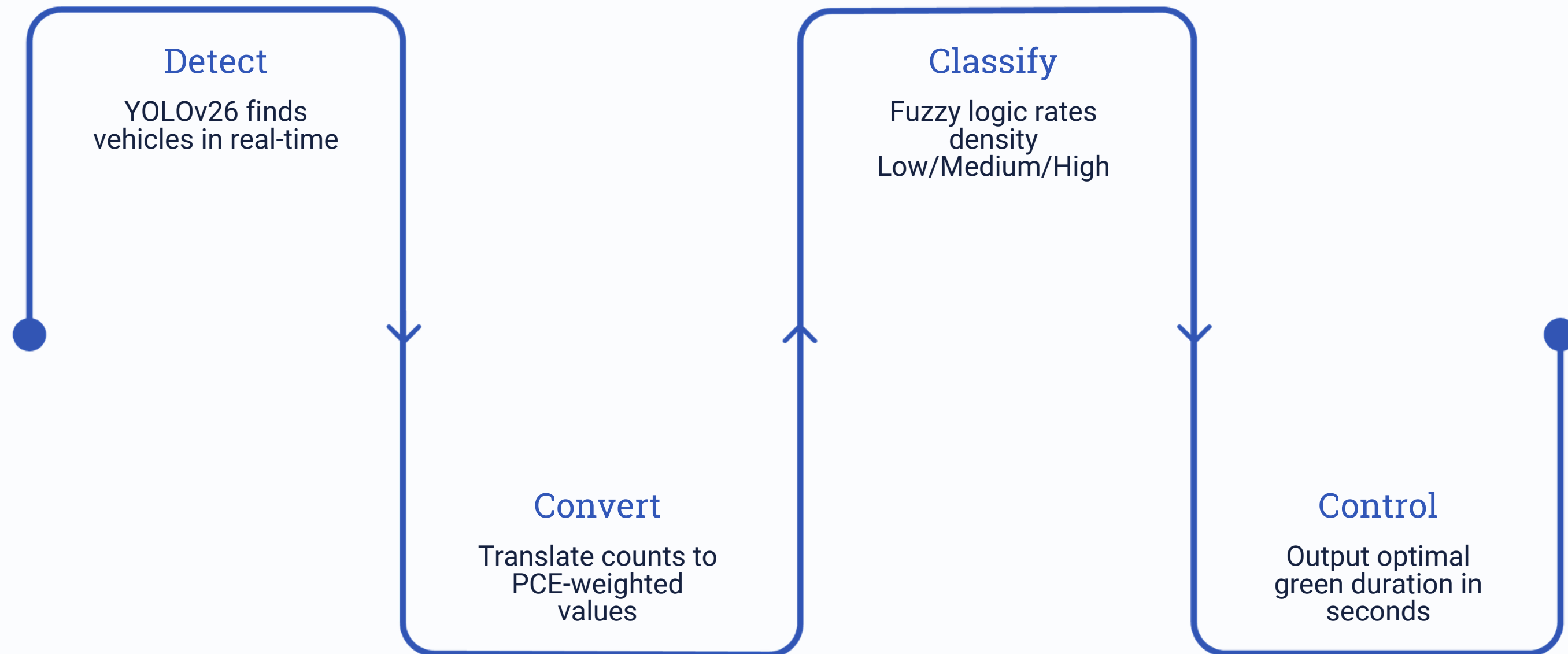
Reinforcement Learning

Computationally intensive and **not yet field-deployed**; fuzzy logic is more stable and interpretable for agencies.

Research Gap

Most prior work uses **older YOLO versions** and idealized Western datasets — not heterogeneous Asian traffic conditions.

Our Solution: YOLOv26 + PCE-Weighted Fuzzy Logic



YOLOv26 is the **latest evolution** in the YOLO series — engineered for edge devices with streamlined architecture, removing unnecessary complexity while delivering faster, lighter deployment. Fuzzy logic handles uncertainty by treating congestion as a **spectrum**, not just empty or full, and dynamically scales green time based on actual demand.

Dataset & Exploratory Data Analysis

Dataset Overview

Open-source Indonesian vehicle dataset from **Roboflow**
(**Arkhsat Project**)



10,500+

Total Images

5 Classes

Pickup, Bus, Car, Motorcycle, Truck

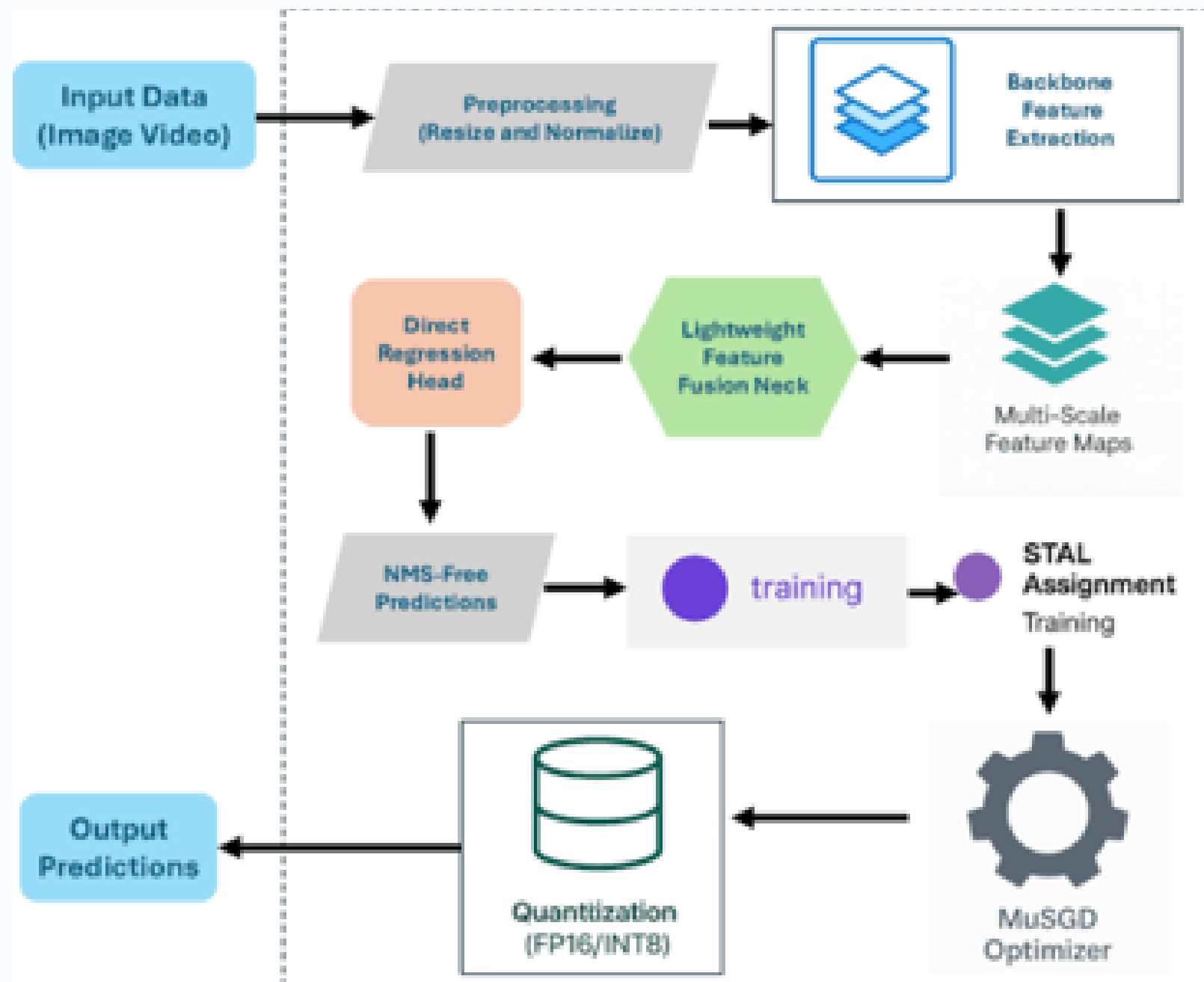
HD

High Definition JPEG

EDA Findings

Unlike traditional traffic repositories with severe class imbalances, this dataset maintains a **balanced footprint** — every category (bus, truck, car, motorcycle, pickup) safely exceeds **2,000 instances**. This structural homogeneity avoids majority-class bias in YOLOv26, ensuring reliable density data for the fuzzy logic controller.

YOLOv26 Architecture & Preprocessing



Why YOLOv26 Nano?

Best **speed-accuracy trade-off** for edge deployment.

Key components:

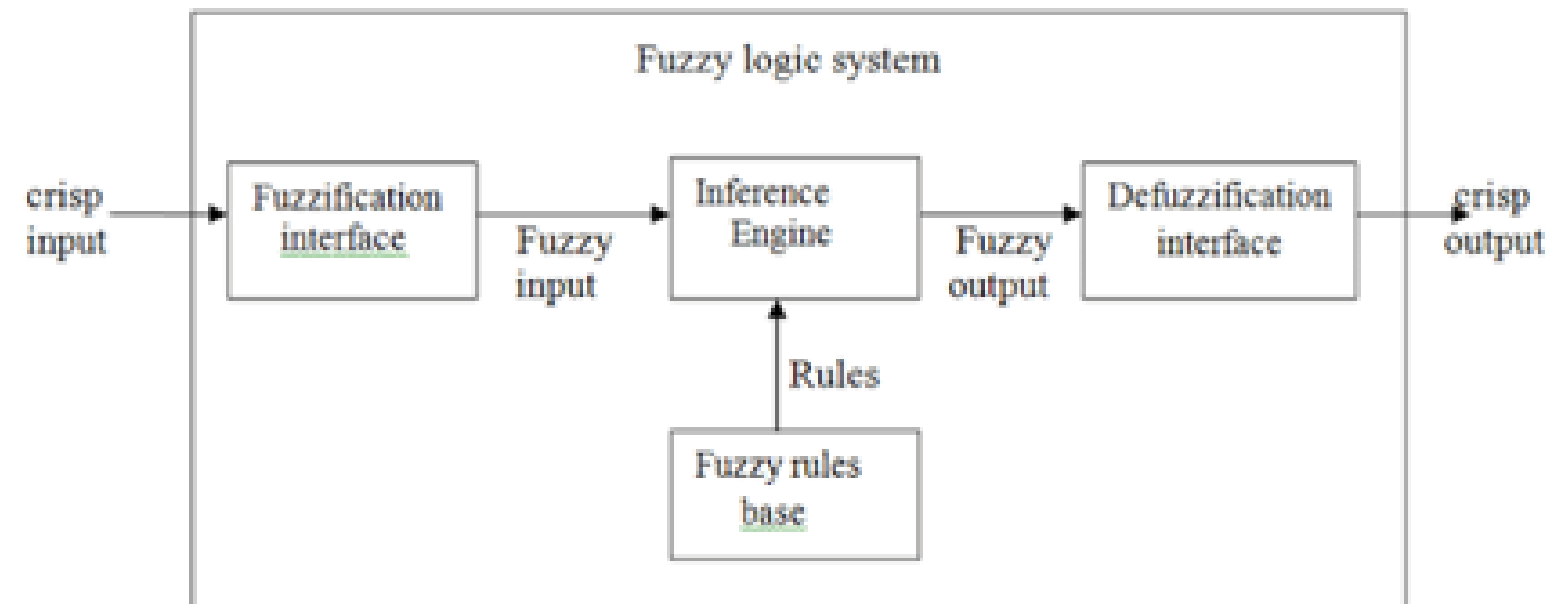
- **Backbone** — feature extraction
- **Feature Fusion Neck** — multi-scale learning
- **Direct Regression Head** — efficient prediction
- **STAL assignment + MuSGD optimizer** — boosted training
- **FP16/INT8 quantization** — faster real-time inference

Preprocessing is **automated** by Ultralytics: letterbox padding, mosaic stitching, horizontal flips, and color modifications simulate varied road conditions and occlusion scenarios.

PCE-Weighted Fuzzy Logic Controller

Passenger Car Equivalent (PCE)

Assigns weights per vehicle type (MKJI 1997 standard) for accurate congestion measurement in heterogeneous Indonesian traffic:



Fuzzy Logic Architecture

01

Fuzzification

PCE density → fuzzy values (Low/Medium/High)

02

Inference

Apply rule base to fuzzy inputs

03

Rule Evaluation

Determine applicable green duration rules

04

Defuzzification

Output crisp green light duration (seconds)

Motorcycle
0.25

Light Vehicle
1.0

Heavy Vehicle
1.3

Model Comparison & Detection Performance

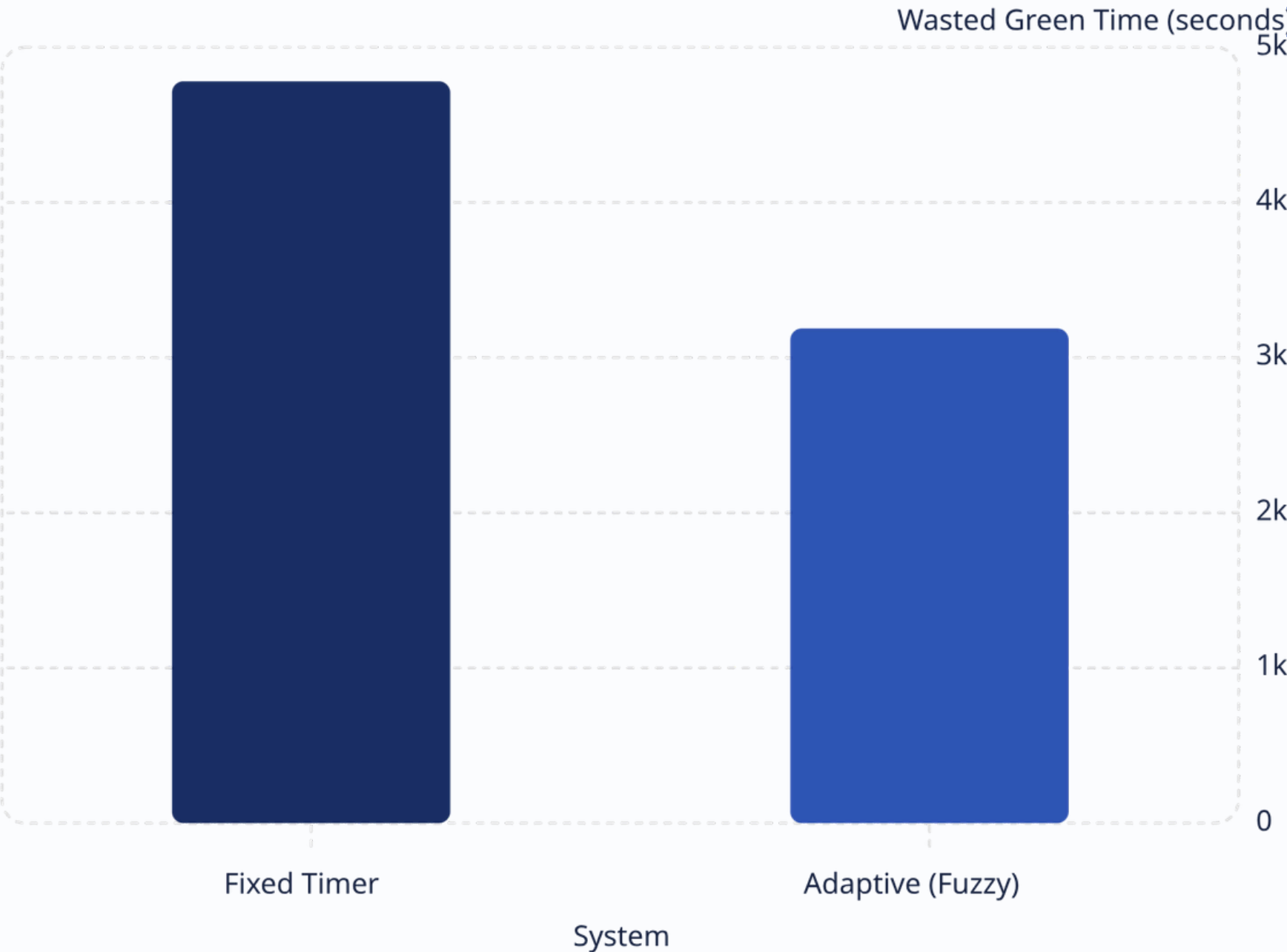
Comprehensive Metric Comparison

Model	Precision	Recall	F1-score	mAP@0.50	mAP@0.50:0.95
YOLOv26n	0.9614	0.9440	0.9525	0.9792	0.8113
YOLOv8n	0.9477	0.9308	0.9392	0.9714	0.8033
YOLOv10n	0.9586	0.9381	0.9482	0.9746	0.8079
YOLOv11n	0.9582	0.9453	0.9517	0.9758	0.8081

Why YOLOv26n Won

Highest precision (0.9614) and best localization accuracy (mAP@0.50:0.95 = **0.8113**) among all compared models. While YOLOv11n achieved the highest recall (0.9453) and F1-score (0.9517), **YOLOv26n was selected** for its superior overall accuracy — critical for precise traffic density estimation and adaptive signal control.

Adaptive System vs. Fixed Timer



Key Findings

33.3% Reduction
Wasted green time dropped from **4,781s** → **3,188s**

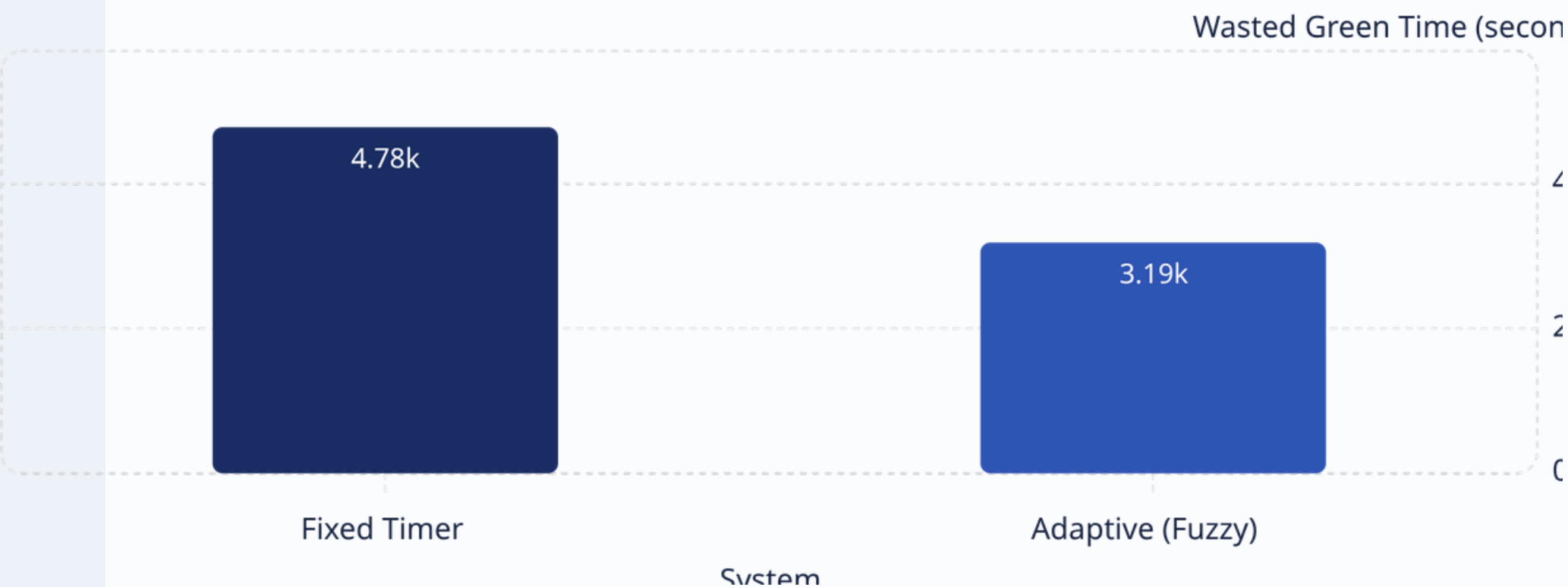
Dynamic Scaling
Fuzzy system concentrates green phases around **20–21 seconds**, cutting off as demand declines

Edge Latency: 3.2ms
Outperforms YOLOv8n (8.5ms) and Faster R-CNN (45.0ms) on edge devices

Throughput Trade-off
YOLOv26n: 80.1 FPS on Tesla T4 — slower than YOLOv8n (115.5 FPS) but highest detection accuracy

Wasted Green Time Reduction

Column chart comparing Fixed Timer (4,781 seconds) vs Adaptive Fuzzy (3,188 seconds) wasted green time.



Key findings:

33.3% Reduction

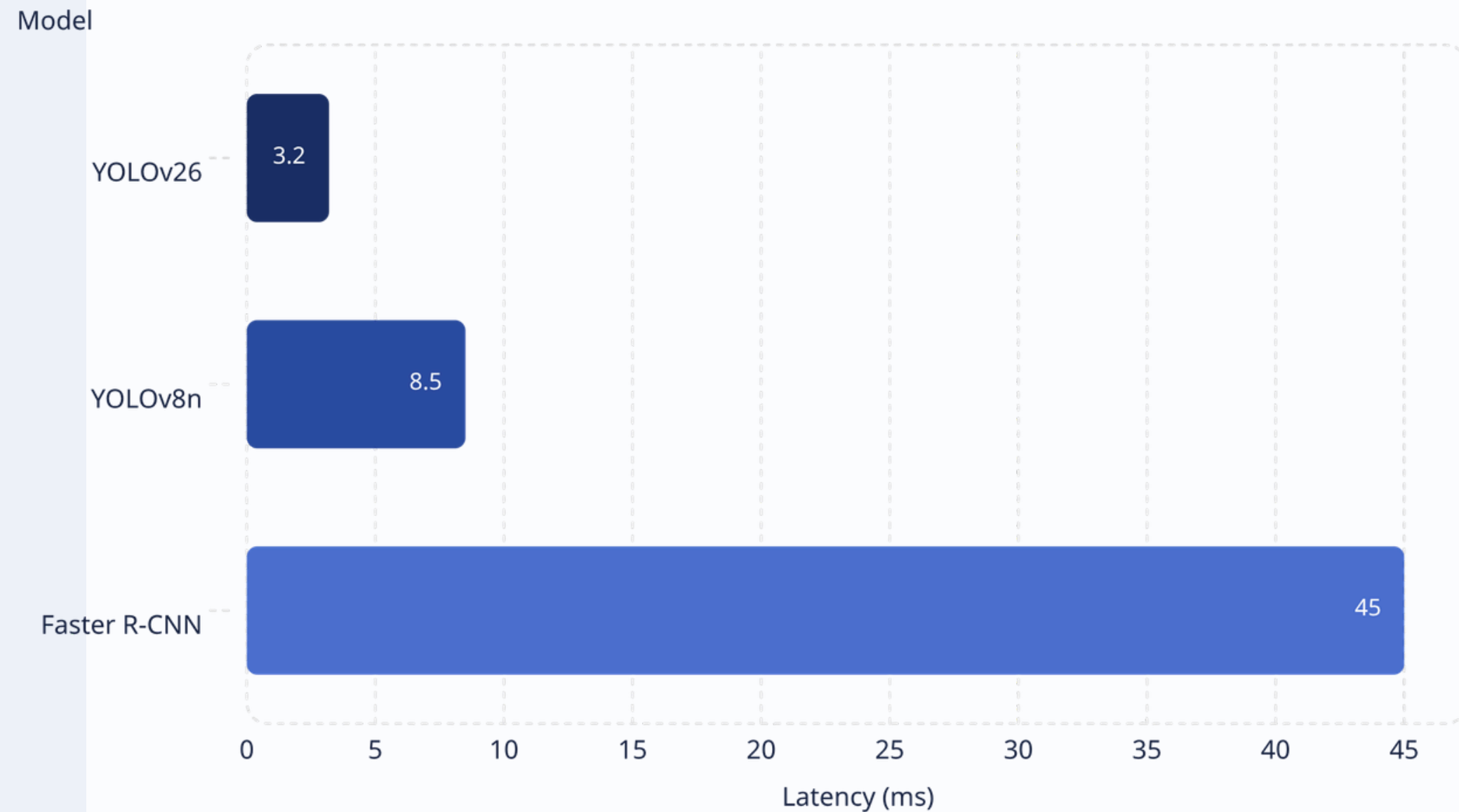
Adaptive control cut wasted green time by nearly a third.

Adaptive Advantage

Green time was used more efficiently, reducing idle periods and improving responsiveness.

RESULTS

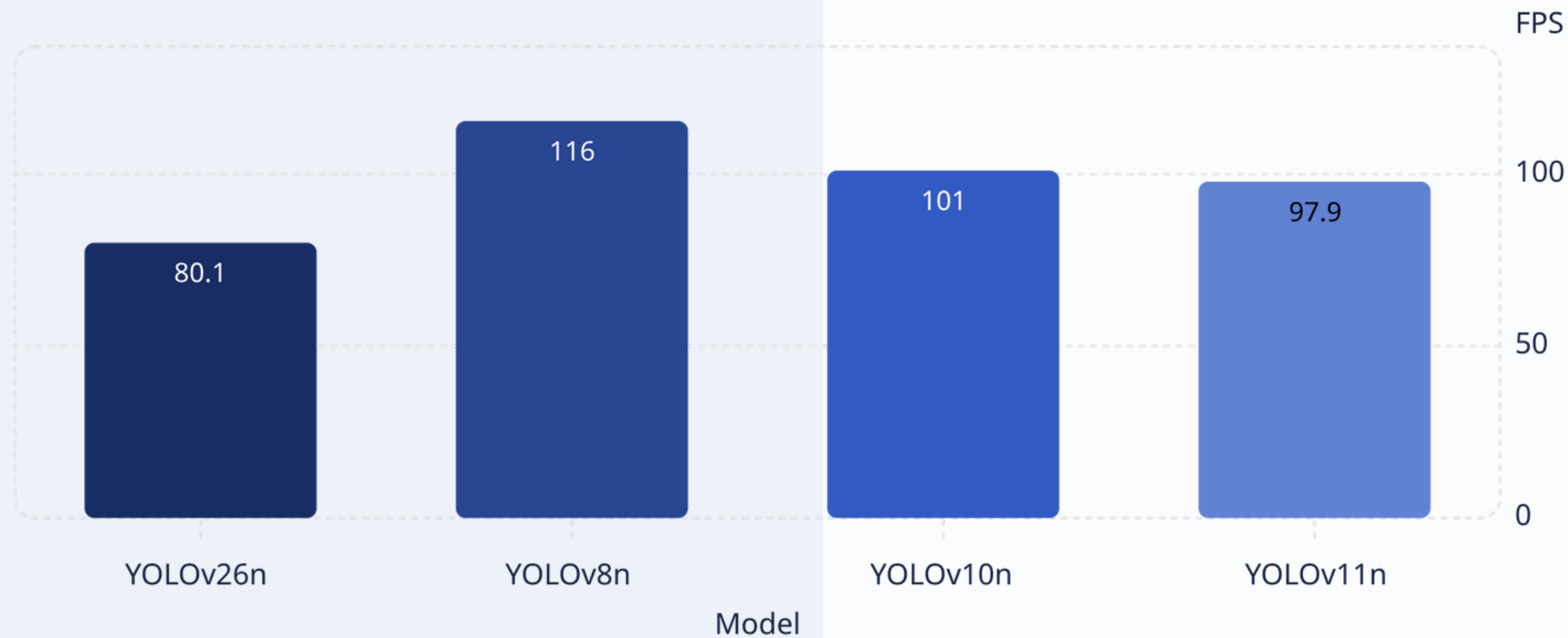
Inference Latency on Edge Device



YOLOv26 is the fastest model at **3.2 ms**, making it the best fit for real-time edge control. That speed is critical for reacting quickly to live detector updates and keeping traffic signal decisions current.

RESULTS

Model Throughput Comparison on Tesla T4 GPU



YOLOv26n processes **80.1 frames per second**, which is slower than YOLOv8n at **115.5 FPS**. However, for traffic control, accuracy is more important than speed. Better detection quality means better traffic decisions.

Conclusion & Future Work

What We Achieved

- YOLOv26n achieved **0.9614 precision** and **0.8113 mAP@0.50:0.95** on Indonesian vehicle dataset
- PCE-weighted fuzzy logic reduced wasted green time by **33.3%** vs. fixed timers
- Edge inference at **3.2ms** — suitable for real-time deployment
- Scalable, responsive framework for **smart city congestion management**

Limitations & Future Work

- **Real-World Deployment**
Validate on actual urban infrastructure beyond simulation
- **Multi-Intersection Sync**
Coordinate signals across multiple intersections
- **Additional Sensors**
Integrate radar and thermal cameras for all-weather robustness
- **CAV Integration**
Enhance fuzzy logic for connected and autonomous vehicles

Thank You

We appreciate your time and attention. Here are some resources for further exploration.

[Code Link](#)

[Dataset](#)